

A Comparative Study of Deep Learning Models for Image Tampering Detection Using CASIA V2.0 Dataset

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Abstract—Detecting images that have been edited or otherwise altered can be a complex task because so many different tools are available to make these edits, and it is becoming increasingly difficult to determine what is real. In this study, we evaluated a number of different deep learning architectures, including ResNet-50, EfficientNet-B4, Vision Transformer, Swin Transformer, ConvNeXt, CAT-Net, and ManTra-Net, using the CASIA V2.0 dataset in order to compare the accuracy with which each architecture was able to differentiate between fake and real images. All of the models were trained using identical methods and were evaluated using multiple metrics (i.e., accuracy, precision, recall, F1 Score, area under the ROC curve (AUC), and misclassification). The Swin Transformer achieved the highest score, with an accuracy of 86.96% and an AUC of 0.9589, demonstrating that it can identify not only large, obvious edits but also small, subtle edits in images. While the ManTra-Net model did not achieve an accuracy level comparable to that of the Swin Transformer, it did require substantially less computational power than the other architectures, which would be advantageous in situations where resources are limited. Overall, these results serve to underscore the critical role of powerful deep learning models in helping identify fake from real images.

Keywords— Image tampering detection, deep learning, convolutional neural networks, vision transformer, Swin transformer, CASIA V2.0 dataset, digital image forensics

I. INTRODUCTION

Advancements in technology have led to an increase in digital image editing tools that can easily alter pictures. These advancements have raised questions and concerns regarding the authenticity and reliability of the images. Techniques that alter the content of an image, such as splicing, copy-move forgery, and object removal, can easily alter the image without visible indicators of forgery. Therefore, it is crucial to develop tools that verify the authenticity of digital images since this is increasingly becoming a challenging task, especially in areas such as journalism, criminal investigations, medical images, surveillance, and digital forensics.

Currently, conventional image forgery detection algorithms depend on features created manually. Some of these include inconsistencies in noise, compression artifacts, lighting alterations, and discontinuity on edges. Such techniques can efficiently identify tampered images in laboratory environments; however, they cannot handle other types of tampering and real-world conditions. In this research, deep learning approaches will be applied in order to automate

the image tampering detection process. Implementation and evaluation of the top-performing convolutional neural networks (CNNs) and transformers will be done based on the CASIA V2.0 image database.

The following lists contributions from this research:

- 1) The rigorous implementation and performance evaluation of several deep-learning models for detecting image forgery that were tested using the CASIA V2.0 database across multiple types of tampering.
- 2) The systematic comparison of convolutional neural networks, transformers, and hybrid models to establish whether there were significant differences in reliability among these three types of architectures when evaluated using a controlled experiment.
- 3) Detailed performance comparisons based on standard evaluation metrics (i.e., accuracy, precision, recall, F1 score, area under the curve [AUC], misclassification error) to assess the quality of classification from the models developed; and,
- 4) The analysis of computational complexity (in terms of floating-point operations, number of parameters, and time to infer from the model) to examine tradeoffs associated with model performance, implementation, and efficiency.

II. LITERATURE REVIEW

Detecting forgeries of digital images has turned out to be an important area of study in the field of digital forensics because of the availability of more sophisticated image editing tools, as well as improper use of edited images via numerous digital media platforms. Inability to modify an image in such a way that there are no alterations visible has resulted in several important questions regarding image validity.

In terms of detecting forged images, early studies involved the development of techniques based on machine learning methods using manually defined features. It was believed that forgeries would lead to statistical irregularities like the presence of noise, variations in illumination, artifacts from the use of color filter arrays, inconsistency in edge regions, and compression artifacts. Although these handcrafted techniques have shown promising performance during testing, they lacked adaptability to real forgeries.

The recent developments in the field of deep learning

technology have revolutionized the process of image forgery detection. With the help of advanced deep learning techniques, it is possible to automatically hierarchically extract features from raw image data. The introduction of CNN models has brought a paradigm shift in the area of forgery detection through the effective process of representation learning. The ResNet model made use of residual learning, which has enabled the training of deeper neural networks without falling prey to the phenomenon of vanishing gradients. The EfficientNet technique furthered the cause of effective forgery detection using a compound scaling technique.

A major breakthrough in the field of computer vision with the help of transformer models has been seen recently. These models utilize the concept of self-attention in order to effectively learn global features. The Vision Transformer model breaks down the image into patches, and the long dependencies between the features are captured by means of self-attention. The idea behind the Swin Transformer model is to introduce the concept of hierarchical feature representation along with shifting window self-attention.

Advancements in deep learning techniques are a game-changer in the area of forgery detection on images. Deep learning techniques are capable of automatically discovering the features of the image data in multiple layers. It is observed that CNN-based models are highly efficient in forgery detection and provide an accurate spatial and semantic representation. ResNet uses residual learning, which enables constructing deep neural networks without any problem related to vanishing gradients. EfficientNet enhanced the performance and efficiency of deep learning models using the compound scaling technique.

Recently, self-attention mechanisms used in transformers have demonstrated high effectiveness in the field of computer vision. Vision Transformer decomposes an image into a set of patches and uses a multi-head self-attention mechanism for building a long-term dependency between patches. Swin Transformers are based on different scales due to the hierarchical structure of the feature and shifted window self-attention. Therefore, transformer-based models are the best choice in forgery detection problems because of their characteristics.

III. METHODOLOGY

In this section, the overall methodology employed for the task of image tampering detection will be explained. The goal is to provide a consistent methodology that allows for a fair comparison among various deep learning models by maintaining similar experimental conditions for all the experiments conducted in this project. In this regard, a description of the datasets, data preprocessing, model architecture, and methodology for evaluation will be provided for image tampering detection.

A. Framework

The proposed solution framework includes five main phases: dataset collection, data preprocessing, training of the deep learning model, predictions using the trained model, and performance analysis of the model. Initially, data was collected from the CASIA V2.0 dataset, which was divided into two main categories: authentic and tampered images. The collected images were then pre-processed and resized

according to the input requirements of various deep learning algorithms. Finally, multiple deep learning models, such as CNNs and transformers, were trained on the dataset for the task of image classification.

The entire process could be summarized as follows:

CASIA V2.0 Dataset → Data Preprocessing → Model Training → Prediction → Performance Evaluation

B. Dataset

All experiments were performed on the CASIA V2.0 dataset of image tampering detection. This dataset has been extensively used by researchers for developing and benchmarking methods in digital forensics. The dataset comprises authentic as well as tampered images produced through different methods of tampering, like image splicing and copy-move forgery. The authentic images constitute original and unmanipulated images, while the tampered images represent the modified images with tampered visual content. The images in the dataset are diverse in terms of scenes, object classes, illumination, and tampering techniques.

In the experimentation process, the entire dataset was split in an eighty-to-twenty ratio for training and testing purposes, respectively.

C. Data preprocessing

Data preprocessing represents one of the most important steps in the development of the proposed image manipulation detection algorithm since the quality and uniformity of input data have a crucial impact on the capabilities of feature extraction, optimization processes, and the general model performance. Because the CASIA V2.0 dataset includes images that differ with respect to spatial resolution, illumination, background, object distribution, as well as various manipulations performed on them, the entire data preprocessing procedure is necessary to facilitate consistent results from different deep learning architectures. Firstly, the CASIA V2.0 dataset was separated into two major classes: genuine and manipulated images. Then, the entire dataset was split into training and testing sets by applying the eighty-to-twenty rule. In addition, all experiments conducted within this study have been performed after implementing the entire data preprocessing procedure under a single configuration.

- **Image Resizing and Spatial Standardization:** The images within the CASIA V2.0 dataset vary widely in dimensions and proportions, such that they cannot be used in the training of neural networks with a specific fixed size. Thus, all images were processed as per the input requirement of each particular architecture used. ResNet-50, Vision Transformer, Swin Transformer, ConvNeXt, CAT-Net, and Mantra-Net use an input dimension of 224 x 224 pixels. Conversely, the EfficientNet-B4 uses an enhanced approach, using an input dimension of 380 by 380 pixels, owing to the fact that the latter allows for preservation of fine manipulation artifacts and forensic traces. Spatial standardization ensures that there is consistency in feature extraction and calculation in all architectures.

- **Data Augmentation:** To increase generalization and avoid overfitting of the models, data augmentation methods were employed during the training process. The application of random flipping on images helped achieve variations in visual appearance while maintaining semantically soundness. Through data augmentation, the models were made robust to different image transformations and variations because they

learned manipulation-invariant features from multiple viewpoints.

- **Normalization and Scaling:** Each image underwent normalization considering the average and dispersion measurements according to the ImageNet database. Image normalization includes changing the distribution of pixels to a standardized structure, which helps in reducing variable changes within an image and makes the gradient descent more efficient. As a result, training processes become faster due to more efficient calculations when implemented in deep neural networks.

- **Tensor Representation:** Images undergo representation in tensor form using the PyTorch deep learning framework. Tensors include data representation through numerical measures in several dimensions. It is necessary since it provides efficient execution of parallel computations within a GPU setting when used during the training and inference processes.

- **Batch-wise Data Loading:** Batch-wise loading of preprocessed data was implemented using the DataLoader method offered by PyTorch. In this way, it becomes possible to perform effective handling and processing of data in batches. Batch-wise loading of image files: Sequential loading of preprocessed data was implemented using the DataLoader class provided by PyTorch.

- **Unified Preprocessing Framework:** A single method was employed regarding the preprocessing of all architectures to ensure that an objective analysis can be performed. By maintaining consistency in preprocessing, one can analyze the impact of the architecture designs, irrespective of their preprocessing techniques, as the conditions are identical for all three different architectures.

D. Model Architectures

A comparative analysis of state-of-the-art image tampering detection methods requires evaluation of multiple deep learning architectures, which incorporate various paradigms of convolutional neural networks, transformers, attention mechanisms, and specialized forgery detection approaches. For this purpose, several architectures representing each category and providing good performance, scalability, and applicability to modern computer vision tasks were chosen based on their structural properties. All the selected models were modified for binary image classification by changing the structure of the classification layer to detect tampered and genuine images. The use of transfer learning, where ImageNet weights were adopted for better generalization ability and faster training process, was also applied since the amount of available forensic training data was relatively small.

1) ResNet-50

ResNet-50 is a deep convolutional neural network architecture built on the residual learning principle. Residual skip connections were used within this network structure to ensure fast backpropagation and minimize gradient problems that typically occur in deep neural networks. Learning residual mappings instead of mapping functions allows for the efficient training process of deep architectures while retaining spatial information. ResNet-50 was applied in the work to serve as a benchmark for a typical convolutional neural network approach to image tampering detection. This model can extract visual features, such as texture features, edge features,

and manipulation characteristics in the manipulated images from the CASIA V2.0 dataset.

2) EfficientNet-B4

EfficientNet-B4 is a state-of-the-art convolutional neural network architecture that proposes compound scaling that aims to achieve a more optimal balance among network depth, width, and input image size during scaling processes compared with existing models. In contrast with the previous scaling methods, the architecture optimizes both the network and resolution simultaneously and thus improves feature extraction efficiency at a lower cost. In this paper, EfficientNet-B4 adopts larger images with a size of 380×380 to extract fine forensic information of the manipulated images.

3) Vision Transformer

Vision Transformer is an architecture that utilizes the idea of applying the self-attention concept to the analysis of image patches in order to learn features. As opposed to other methods that use convolutions, this architecture splits the input image into patches and further treats these patches as sequence tokens using transformer encoder layers. The self-attention process allows capturing long-range dependencies and global dependencies between different parts of an image. This architecture was chosen in the current study for evaluation purposes in relation to its capabilities of recognizing global manipulation inconsistencies and semantic artifacts in manipulated images of the CASIA V2.0 database.

4) Swin Transformer

Swin Transformer is a vision transformer architecture that aims at improving computational efficiency and the capability of modeling local features by means of shifted window attention. This architecture implements shifted windows during the process of performing self-attention in order to facilitate information exchange between different windows. In the performed experiments, this architecture outperformed other architectures in terms of overall metrics and proved to be the most efficient in detecting manipulation-related image structure and semantic artifacts.

5) ConvNeXt

ConvNeXt is an advanced version of the convolutional neural network, which has been inspired by some common architectural choices seen in transformers. These choices include the use of large convolution filters, layer normalization, inverted bottlenecks, and architectural tuning to boost its efficiency.

For this study, ConvNeXt has been employed for analyzing if transformer-inspired convolutional networks can match the performance of other types of networks in detecting tampered images.

E. Training Architecture

All deep learning architecture adopted in this study was trained using the PyTorch library within the Graphics Processing Unit environment to enhance the performance of training. The same training environment was adopted by all the models used within the experiment for consistency.

Transfer Learning was done using a pretrained network having ImageNet weights as the initial parameters. This helped speed up the training process and also improved the generalizability of features extracted for detecting tampering

in the CASIA V2.0 dataset. Modification was made to the last classification layer of all models for the binary classification task of authentic and tampered images.

The use of Adam optimizer was important because of its ability to adaptively update the learning rates as the training progresses. This also ensures convergence of the model. The CrossEntropyLoss was used for the classification of labels.

Thirty epochs were used for training all the models under the same training parameters. A learning rate of 0.0001 and batch sizes of 16 and 32 were considered according to the complexity of each network.

The main hyperparameters that were used in this experiment are listed below in the table:

Hyperparameter	Value	Justification
Optimizer	Adam	Supports adaptive learning and achieves stable convergence when training DNNs
Learning Rate	0.0001	Provides stability in optimization and avoids sudden changes in the parameters
Batch Size	16 / 32	Selected based on the complexity of the model and the GPU memory available
No. of Epochs	30	Facilitates adequate learning of the model and minimizes overfitting
Loss Function	CrossEntropyLoss	Ideal for binary image classification problems
Input Resolution	$224 \times 224 / 380 \times 380$	Selected based on the architecture requirements for the inputs

Hyperparameter	Value	Justification
Framework	PyTorch	Supports efficient implementation of deep learning with GPU acceleration
Hardware Environment	GPU-Based Training	Accelerates the training and inference process.

This allows us to get an understanding of how good the models are at detecting forgeries as well how heavy they are computationally.

F. Evaluation Metrics

The assessment of the deep learning models was carried out on two fronts: classification-based measures and efficiency measures, thereby offering a holistic evaluation of both the predictive performance and practical applicability of the models on image tampering detection operations. The reason for assessing both aspects is that forgery detection mechanisms need to attain not only high classification accuracies but also efficiency for actual implementation.

The classification me are used to assess the ability of the model to accurately distinguish between genuine and manipulated images using the CASIA V2.0 database. This measure gives a measure of how well the models predict, and this is a binary classification setting.

Classification Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- Area Under Curve
- Misclassification Error

Computational Efficiency Metrics

- Number of Parameters
- Floating Point Operations
- Inference Time

They offer a comprehensive overview of the detection capabilities as well as the computational complexity of the developed models.

IV. RESULTS AND DISCUSSION

The experimental results obtained in the evaluation of multiple deep learning architectures against the CASIA V2.0 dataset via a comparative analysis are presented in this section. These models have been analyzed based on different metrics in order to evaluate their performance, robustness, and suitability in forgery detection applications. The metrics used include accuracy, precision, recall, F1 score, area under the curve, and misclassification error.

A. Model Performance Comparison

The performance of the implemented deep-learning architectures in classifying images from the CASIA V2.0 dataset was evaluated using standard performance metrics, including accuracy, precision, recall, F1 score, area under the curve and misclassification error. These metrics provide an overall understanding of how well each model is able to classify an authentic image from a tampered image, as well as how robust and reliable each model is in doing so. All architectures were evaluated using a unified experimental setup in order to maintain an equal opportunity for comparative analysis. The results presented below clearly show that both transformer-based and CNN-based models can be effectively used in forgery detection.

TABLE II contains a summary of the classification results for each of the implemented models on the CASIA V2.0 dataset.

Model	Accuracy	Precision	Recall	F1 Score	AUC	MCE
ResNet-50	0.7768	0.7019	0.7834	0.7404	0.8433	0.2231
EfficientNet-B4	0.7448	0.6903	0.6741	0.6821	0.8411	0.2552
Vision Transformer	0.7680	0.6767	0.8049	0.7351	0.8160	0.2319
Swin Transformer	0.8696	0.8385	0.8410	0.8397	0.9599	0.1304
ConvNeXt	0.7273	0.6653	0.6614	0.6634	0.8090	0.2727

From the experimental results, it is clear that the Swin Transformer performed much better than other models in accuracy with 86.96% as well as an AUC value of 0.9589. The advantage of Swin Transformer with respect to its performance attributes is because of its hierarchical feature representation and shifted window attention in capturing both local manipulation and global inconsistency in tampered images.

The performance of the ResNet-50 model is quite impressive due to its residual learning ability and hierarchical feature extraction mechanism, which results in good performance with reliable recall and F1 values.

The Vision Transformer model achieved excellent recall performance by efficiently modeling dependencies through the use of a self-attention mechanism. However, its overall

classification accuracy did not match that of Swin Transformer since it cannot effectively model local features as opposed to the latter, that utilizes hierarchical transformers.

EfficientNet-B4 achieved balanced classification performance in the absence of high computational complexity since it effectively utilizes compound scaling for feature extraction and efficient classification performance.

ConvNeXt achieved moderately good classification performance since it adopts a transformer architecture design in a convolutional setting. Its classification performance surpassed that of traditional convolutional networks but was not comparable to that of Swin Transformers.

However, the general conclusion from this experimental study is that the use of architectures based on transformers, especially the Swin Transformer architecture, provides excellent performance in tasks related to image forensics because they can simultaneously capture both local forensic features and global contexts.

B. Discussion

First of all, as the comparative assessment has shown, the choice of an appropriate architecture is key to achieving high efficiency in image forensics tasks. Thus, while CNN-based architectures excel in recognizing local textures and manipulation traces, transformer-based architectures are better suited for understanding global context by means of self-attention mechanisms.

As well as classification performance, the computational efficiency of the implemented architectures was considered by estimating the number of parameters, floating point operations (FLOPs), and the time required for the model inference.

The computational efficiency comparison for each implemented architecture can be found in Table III.

Model	Parameters	FLOPs	Inference Time (s)
ResNet-50	23.51 M	4.13 G	0.0113
EfficientNet-B4	17.55 M	4.61 G	0.0165
Vision Transformer	85.61 M	16.85 G	0.0189
Swin Transformer	27.81 M	4.46 G	0.0223
ConvNeXt	27.50 M	4.37 G	0.0141

Inferences from the computational analysis reveal a large difference in complexity between the tested architectures. For example, the ResNet-50 model took the least time in inference, implying that the computational efficiency was high enough to classify images in an efficient manner. EfficientNet-B4 attained computational efficiency by making use of fewer parameters as compared to the other models.

The Vision Transformer needed the highest parameters and computations as a result of its global self-attention mechanism and transformer architecture based on patches. However, despite the success that the Vision Transformer demonstrated in efficiently acquiring context-aware features, it brought about a rise in the computational complexity and thus resource consumption.

The most effective combination of detection performance and computational efficiency was achieved by the Swin Transformer due to its shift window attention. As opposed to other transformer-based architectures, the Swin Transformer was noticeably less computationally complex than others.

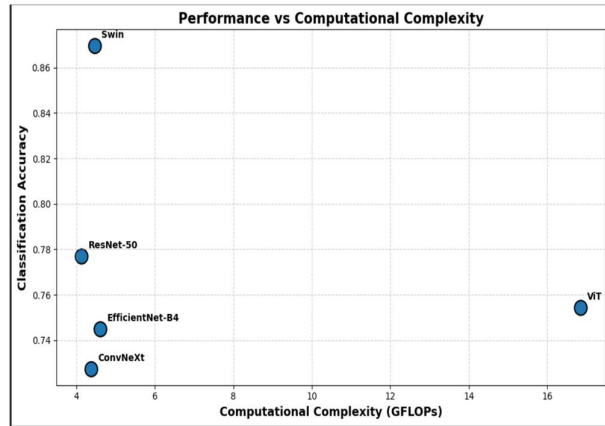


Fig. 1: Scatter plot of computational complexity of models and classification accuracy.

This figure shows the trade-off between the complexity and accuracy of each evaluated model. From these results, we see that the Swin Transformer achieves the best performance while keeping a moderate. Comparatively, Vision Transformer needs more floating-point operations and can boost the performance less.

In regard to the complexity of the models, the ConvNeXt exhibited only moderately high levels of computational complexity but made use of some architectural advancements of transformers in a convolutional framework.

In general, the results show that transformer-based models tend to have greater requirements for computational power in comparison to convolutional neural network-based models. Still, their advanced ability to capture long-range contexts positively influences the tampering detection performance.

Out of all implemented models, Swin Transformer achieved the best tradeoff between classification accuracy and feature extraction capabilities. The hierarchical attention mechanism allows modeling the fine features as well as semantic inconsistency patterns, which play an important role in forgery detection tasks.

This research proves the superiority of modern deep learning models over traditional hand-crafted approaches in terms of automatic learning of manipulation-sensitive features without relying on any prior knowledge. Nonetheless, transformer-based models are computationally more demanding than conventional convolutional neural network models.

These results prove the efficacy of deep learning-based approaches for the task of automated image tampering detection.

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Recent developments in the field of image tampering detection, deepfake analysis, transformer networks, adversarial robustness, and forensic feature learning have resulted in a significant boost in the capabilities of automated forensic systems. Several recent researches have been carried out focusing on making forgery detection models more robust and generalized under real-world conditions. Adversarial robustness studies on face forgery detection revealed that deep learning models were vulnerable to black box attacks and needed secure forensic frameworks [1]. Transformer network-based solutions, such as spatial-frequency dual-stream vision transformers, showed their superiority in manipulation-sensitive representation learning in deepfake detection applications [2]. Additionally, multi-granularity forensic feature consistency analysis was performed for improving the effectiveness of copy-move forgery detection by analyzing local feature correlations [3]. Deep supervised anomaly detection techniques were also developed for improving forgery detection generalization ability [4].

Some of the most recent literature investigated the problem of frequency-domain feature learning along with transformer architectures for capturing very fine-grained forgery clues from images [5], [6]. Dual-stream learning frameworks for handling both temporal and spatial information, as well as phase-based edge features, have achieved outstanding results in the task of forgery detection for cases where spatial inconsistencies along with temporal artifacts occur [7]. Moreover, some papers considered the problem of deepfake generalization for overcoming generalization problems associated with deepfake detection and discussed how to mitigate the problem of spurious correlation [8]. Text manipulation and forensic transformers are two important trends in real-world forgery detection benchmarks [9]. Landmark-guided feature learning techniques showed improved capabilities in deepfake generalization, paying attention to the manipulation of sensitive facial areas [10].

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